Is the Transmission of ECB Monetary Policy Surprises to Euro-Area Sovereign Yields State-Dependent?

Monjish Sarkar

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Summary

The primary objective of this dissertation is to analyse whether ECB monetary policy surprises transmit to euro area sovereign yields in a state-dependent manner. Using high-frequency surprises from the EA-MPD (Euro Area Monetary Policy Database) and sovereign yield responses for Germany, Italy, France and Spain, it separates ECB communication into press release target surprises, which capture unexpected news about the current policy decision, and press conference timing and forward guidance surprises which capture unexpected news about the timing and expected path of policy. The empirical section estimates event study regressions and interaction specifications using VSTOXX, VIX, and debt-to-GDP as state variables. The results indicate that press conference communication is priced into sovereign bond markets, consistent with hawkish ECB communication increasing expected future interest rates. However, transmission is not consistent across different states. While VSTOXX interactions are negative and the VIX robustness reinforces this pattern, debt-to-GDP amplifies the impact in the forward guidance channel. Overall, the dissertation demonstrates how state-dependence depends on the state measured.

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1 Introduction

The transmission of euro area monetary policy depends not only on the actions of the European Central Bank (ECB) but also on how financial markets interpret the information contained and conveyed by the ECB communication. Ever since the Global Financial Crisis, and specifically during the period of unconventional monetary policy, the ECB has relied on communications to shape expectations regarding future path of interest rates and asset purchases.

ECB policy events consist of more than one type of news. First, the press release reveals the nature of the immediate policy decision. Second, the subsequent press conference provides a more detailed explanation regarding policy change and provides information about the timing, persistence, and probable future path of policy. In the context of sovereign bond markets, this distinction is central because yields are forward looking prices, and so they respond to expectations about future short-term rates, inflation, term premia and liquidity conditions along with the current policy decision.

In this dissertation, I examine whether the transmission of ECB monetary policy surprises to euro area sovereign yields is state-dependent. It essentially asks whether the response of sovereign yields to ECB monetary policy news changes when uncertainty is high. The central research question is therefore, “Does uncertainty dampen or amplify the response of euro-area sovereign yields to ECB monetary policy surprises?” This question is relevant because standard event study estimates tend to report an average effect of monetary policy surprises on asset prices. However, an average effect could obscure important variation across market states i.e. a hawkish ECB surprise may have different effects in calm market conditions as compared to more uncertain market conditions.

To address this question, I use high frequency monetary policy surprise data from the Euro Area Monetary Policy Event-Study Database developed by Altavilla et al. (2019). This EA-MPD approach involves the use of high-frequency changes in market interest rate instruments around ECB communication windows to isolate the unexpected component of the ECB policy news. In this dissertation, the press release target surprise which captures the unexpected information about the current policy decision, is used as a preliminary comparison. The main analysis focuses on the press conference timing and forward guidance surprises: the former captures revisions in market expectations while the latter captures revisions to the persistence and expected path of future policy.

The dependent variables are high frequency event window changes in sovereign yields for Germany, France, Italy and Spain. The main analysis uses 10-year sovereign yields, while 2-year and 5-year yields are used in a term-structure extension. By using event window yield changes, the analysis focuses on the immediate reaction of the financial market to ECB policy news rather than on slower low frequency movements in sovereign yields. This high frequency approach is important since it helps with the isolation of market responses to unexpected ECB communication from other macroeconomic developments.

In the empirical analysis, I first estimate the average response of sovereign yields to ECB press conference surprises followed by a test for state-dependence by interacting the ECB surprise factors with measures of market based financial uncertainty and fiscal vulnerability. VSTOXX, a euro-area equity-market implied volatility index, is used as the baseline uncertainty measure because it captures European equity market-based volatility. I also use VIX, a US equity market implied volatility index, often used as a broader global financial volatility measure, as a supporting robustness check. I also include a debt-to-GDP extension to examine whether fiscal vulnerability, or fiscal uncertainty, strengthens the response of sovereign yields to ECB communication.

The baseline results show that ECB press conference communication is relevant for euro area sovereign yields. Both timing and forward guidance surprises raise 10-year sovereign yields across Germany, France, Italy and Spain. This is consistent with the expectations theory of the term structure, under which long term yields reflect expectations about future short-term interest rates as well as term premia. A hawkish press conference surprise raises expectations about the future policy path, making existing fixed coupon bonds less attractive. Therefore, bond prices fall, and yields rise.

The state-dependent results show that the response of sovereign yields to ECB communication is weaker when market uncertainty is higher. The VSTOXX interaction terms are negative across the individual country by country regressions, especially timing surprises, indicating that higher uncertainty dampens the yield response to ECB press conference shocks. The VIX robustness results point in the same direction, suggesting that this dampening result is not specific to the euro area VSTOXX measure. Importantly, the interpretation is not that uncertainty mechanically lowers sovereign yields, rather, the interaction terms show that the marginal response of sovereign yields to ECB communication is smaller when uncertainty is higher.

The debt-to-GDP extension points to a different mechanism. Market uncertainty dampens the immediate response to ECB communication, whereas fiscal vulnerability may amplify the response to forward guidance surprises. This distinction is central as it suggests that state dependence does not always work in one direction. Broad market uncertainty and country level fiscal vulnerability therefore capture different dimensions of uncertainty and so the response need not be uniform across state variables.

This dissertation contributes to the existing literature in three distinct ways. First, it builds on the high frequency monetary policy surprises literature by using the EA-MPD high frequency event-study approach to the question of state-dependent sovereign yield transmission. Second, it highlights a clear difference in observed mechanisms by using either the uncertainty proxies or fiscal vulnerability measures rather than forcing all state dependence into a single mechanism.

Third, it allows the analysis to assess both the cross country and time varying patterns in ECB transmission by comparing country level results to pooled panel estimates and period split specifications.

The rest of the dissertation is structured as follows. Section 2 reviews the relevant literature and develops the conceptual framework linking ECB communication, uncertainty and sovereign yields. Section 3 describes the data and variables, including the EA-MPD surprise factors, sovereign-yield responses, VSTOXX, VIX, and debt-to-GDP ratios. Section 4 presents the empirical methodology, including the baseline event-study regressions, interaction specifications, pooled panel model, and period splits. Section 5 reports and interprets the empirical results. Section 6 concludes by summarising the main findings, discussing limitations, and outlining possible extensions.

2 Literature Review and Conceptual Framework

In this section, I review the literature relevant to this dissertation's research question through three primary strands. The first concerns the measurement of monetary policy surprises through high frequency event-study methods. The second is used to focus on the ECB communication and to highlight the distinction between the press release and the press conference parts of the communication. The third strand examines uncertainty, financial stress, and state-dependent monetary policy transmission while linking sovereign yields to fiscal vulnerability and risk premia. Together they motivate the empirical question of whether ECB monetary policy surprises have stable effects on sovereign yield changes and whether they are state dependent on the prevailing market and fiscal environment in the euro area.

2.1 High-Frequency Monetary Policy Surprise Literature

Identification is the central empirical challenge in monetary policy research. Policy decisions are not random; central banks respond to macroeconomic conditions, and financial markets often anticipate policy decisions before they come into effect. As a result, if a researcher compares interest rates or asset prices before and after monetary policy changes over low frequency data, then the separation of the effect of monetary policy from the macroeconomic news and expectations could become complicated. High-frequency studies solve this problem by focusing on narrow time windows around central bank announcements. The logic is that changes in asset prices within a short window are more probable to reflect unexpected policy news rather than broader macroeconomic developments.

This high-frequency approach has been influential in United States literature with Kuttner (2001) using the federal funds futures to separate anticipated and unanticipated components of Federal Reserve policy decisions. The main point is that only the unexpected component should influence financial markets during announcements. Gürkaynak, Sack and Swanson (2005) extended this approach by explaining that announcements regarding monetary policy contain more than one type of news. They distinguish between a target factor (press release), associated with current policy rate surprises, and a path factor (press conference forward guidance channel) associated with expectations regarding the future path of policy. This is an important distinction because monetary policy works through changes in both current short-term interest rates and communication about future rates.

Later studies also apply the high frequency policy surprise to broader macro-financial questions. In a study, Gertler and Karadi (2015) used high frequency surprises as external instruments to identify monetary policy shocks in a structural VAR framework. Nakamura and Steinsson (2018) in their paper, show that monetary policy announcements contain information regarding central bank's assessment of the economy which gives rise to a "central bank information effect". This is important because a market reaction to a central bank announcement may not only reflect a pure policy shock but also information about growth and inflation as revealed by the central bank. Jarociński and Karadi (2020) further distinguished between monetary policy shocks and central bank information shocks, showing that the interpretation of high frequency surprises depends on the joint movements of interest rates and stock prices.

This literature can be considered the methodological foundation of this dissertation. The unique feature of the high frequency identification is that it isolates the reactions of the financial markets specifically due to the unexpected policy news. The literature also demonstrates how monetary policy surprises are concerned with the current policy decision, the future path of policy, and the expected persistence of rates as revealed by the central bank. In this dissertation, I build on this framework by testing whether the yield response to those components is stronger or weaker when uncertainty is high.

2.2 ECB Communication and Press Conference Channel

In the second strand, I examine the role of the central bank in shaping market expectations primarily through communication. Monetary policy works not only through current policy actions but also through influencing beliefs about the future path of interest rates along with the central bank's reaction function, and persistence of the policy stance. This became particularly important in the post Global Financial Crisis environment, when forward guidance and other unconventional tools increased the importance of communication as a policy instrument.

Blinder et al. (2008) assert that central bank communication may improve monetary policy effectiveness by making the bank's objectives transparent and their reaction function clearer to the markets. If the investors gain a fundamental understanding of how the central bank is going to respond to inflation, output, and financial stress, then communication can affect asset prices even before the policy changes the interest rate. However, communication can contain different types of information. Hansen and McMahon (2016) demonstrate that central bank statements reveal intentions and information regarding its assessment of the economy and as a result market reactions may incorporate expected policy changes and updated beliefs about the macroeconomic conditions.

The ECB is particularly useful for studying communication because its policy events can be separated into a press release and press conference. Altavilla et al. (2019), one of the fundamental papers for this dissertation, exploit this institutional structure in the Euro Area Monetary Policy Event-Study Database by separating ECB announcements into different high frequency windows. This separation enables us to distinguish between news about the current policy decision and news about the future policy path.

For this dissertation, the press release target surprise is initially considered part of the broader EA-MPD event study structure since it captures unexpected news about ECB policy decision. However, the empirical results indicate that the target surprise channel is weaker and less systematic than the forward guidance channel. Therefore, the analysis is centered around the press conference where sovereign yields are more consistently related to ECB monetary policy surprises. This does not mean that the press release is theoretically irrelevant but that in this sample press conference factors are more informative for examining how uncertainty changes the transmission of ECB monetary policy surprises to marginal changes in sovereign yields.

2.3 Uncertainty and State-Dependent Monetary Policy Transmission

Uncertainty can affect both real economic behavior and financial market pricing. Bloom (2009) shows how uncertainty shocks may reduce investment and hiring by increasing the value of waiting. Baker, Bloom and Davis (2016) develop economic policy uncertainty indices and show that policy uncertainty can be associated with weaker macroeconomic outcomes. Although in this dissertation I mainly use market based volatility measures instead of newspaper-based policy indices, the broader picture is relevant. Uncertainty changes how economic agents and investors respond to new information.

Uncertainty can alter the effect of monetary policy surprises on financial markets through more than one channel. On one side, high uncertainty may make investors more sensitive to policy news if it increases the importance of risk premia, liquidity conditions, and sovereign vulnerability. On the other side, uncertainty may weaken the observed response if markets place greater weight on recession risk, safe-haven demand, or the fact that the central bank's future policy path will be revised due to declining economic growth.

The sign of state dependence is not determined in advance, and this ambiguity is important for this dissertation since the direction of state dependence is treated as an empirical question. This section provides the conceptual motivation for the interaction specifications I use later.

2.4 Sovereign Yields, Euro Area Heterogeneity and Fiscal Vulnerability

In a monetary union, sovereign yields reflect both common monetary policy forces and specific risks attributable to each country. On one hand, ECB communication affects all countries by changing expectations about future interest rates, and on the other, sovereign yields differ across countries due to variations in fiscal fundamentals such as liquidity, and the level of government debt relative to GDP. This means the same ECB surprise does not affect German, Spanish, Italian and French sovereign yield changes in exactly the same way.

The sovereign debt crisis specifically highlighted the importance of these distinctions. Spreads between core (Germany, France) and peripheral countries (Spain, Italy) widened sharply during the crisis, reflecting concerns about debt sustainability and banking sector fragility. The existing literature on the euro area sovereign spreads shows that both financial fundamentals and market uncertainty matter for sovereign risk pricing. De Santis (2014), for example, highlights spillovers and flight-to-liquidity effects during the sovereign debt crisis, while Afonso, Arghyrou and Kontonikas (2015) emphasizes the role of fiscal and macro-financial fundamentals in explaining European sovereign spreads.

This literature is relevant for the core/periphery and debt-to-GDP extensions. If ECB communications raise the expected future rates, the effect could be stronger for countries with higher debt burdens because they face increased refinancing costs and debt servicing pressure. Forward guidance is specifically linked to this mechanism because persistent future interest rate expectations would have greater implications for debt sustainability than a short lived one. Debt-to-GDP is therefore used as a state variable in place of VSTOXX for a later extension.

Changes in sovereign yields do not always appear through spreads. A hawkish ECB surprise may raise German, French, Italian and Spanish yields together by shifting the common expected rate. Even though the level response of yields is economically meaningful, the spread may change only moderately even if core and peripheral yields move in the same direction.

Overall, the literature is suggestive of three important points. ECB communication is central to expectations about future policy; uncertainty may either amplify or dampen monetary policy transmission and finally changes in sovereign yield responses may reflect both European monetary policy effects or country specific fiscal vulnerability. These insights provide the basis for the empirical specifications I use in this dissertation.

3 Data and Variables

In this section, I describe the data sources, variables, sample construction, and data cleaning steps that I used in the empirical analysis. The dataset I use primarily combines ECB monetary policy surprise measures, high-frequency financial market responses, market uncertainty indices, and a fiscal vulnerability variable. The unit of observation is an ECB monetary policy event, and in the panel specifications it is an ECB event-country observation.

3.1 EA-MPD Dataset and ECB Surprise Variables

The main monetary policy data I use for this dissertation is from the Euro Area Monetary Policy Event-Study Database developed by Altavilla et al. (2019). The EA-MPD is a high frequency event-study dataset constructed around ECB monetary policy events. It consists of monetary policy surprise factors and financial market responses measured in narrow windows around ECB announcements. A key feature of the EA-MPD is that it separates ECB events into a press release window and a press conference window. This matters for the data because different surprise variables are associated with different windows. The press release window mainly captures immediate policy announcement. The relevant surprise variable in this window is the target surprise, which measures unexpected news about the current policy decision. The press conference window contains the timing, forward guidance, and QE surprise factors. Timing surprises measure unexpected information about when future policy changes are expected to occur. Forward guidance surprises measure unexpected information about the broader expected path and persistence of future monetary policy. QE surprises are also available, but only for a smaller later sample, so they are treated as supporting evidence rather than as the central explanatory variables. The main independent variables used in the dissertation are therefore the ECB surprise factors: target, timing, and forward guidance. These are numerical surprise factors which are supplied by the EA-MPD, derived from high-frequency changes in market interest rate instruments within the ECB communication window. The target surprise is used for the press release check. Timing and forward guidance are used as the main press conference monetary policy surprise variables. These variables are market-based surprise measures and not manually coded textual variables. The surprise variables are used in the units provided by the EA-MPD dataset, which are based on high-frequency changes in euro area interest rate instruments around ECB event windows. They therefore measure the unexpected market-priced component of ECB communication rather than the full announced policy decision.

Table 3.1: Mapping of ECB Surprise Variables to Communication Windows

ECB communication stage	EA-MPD surprise variable	Interpretation	Role in this dissertation
Press release / initial policy announcement	Target surprise	Unexpected news about the current policy decision, such as the immediate policy-rate decision announced at the start of the ECB event.	Used as a preliminary comparison to examine the initial announcement channel.
Press conference	Timing surprise	Unexpected news about the expected timing of future policy changes. It captures revisions in market expectations about when future ECB policy adjustments are likely to occur.	Main press conference surprise variable.
Press conference	Forward guidance surprise	Unexpected news about the expected future path and persistence of monetary policy. It captures how ECB communication changes expectations about future policy beyond the immediate decision.	Main press conference surprise variable.

3.2 Financial Market Response Variables

The dependent variables are high frequency event window changes in sovereign yields. These are measured around ECB monetary policy events and are not ordinary daily, monthly or annual yield levels. The main countries used are Germany, France, Italy and Spain. The main dependent variables are the 10-year sovereign-yield responses:

- Germany 10-year yield response
- France 10-year yield response
- Italy 10-year yield response
- Spain 10-year yield response

The 10-year yield is used as the main maturity because it is a standard benchmark for sovereign borrowing costs. Along with this, these bonds are also highly liquid which is beneficial for the event study design because more liquid sovereign bond markets tend to incorporate ECB communication quickly within narrow windows.

I also extended this analysis to 2-year and 5-year sovereign yield responses. These additional maturities help determine whether the same pattern appears at different points of the yield curve. The dataset also contains other financial market response variables, including broad European equity responses and exchange rate responses. In this dissertation, the sovereign yield variables are the main dependent variables. Equity market responses are used only as supplementary checks and are not the main object of the analysis. For the core/periphery part of the analysis, Germany and France are classified as core sovereign markets, while Italy and Spain are classified as peripheral sovereign markets. This is an operational grouping used to organise the data. Individual country variables are still retained because the four sovereign markets are not identical.

3.3 Market uncertainty variables: VSTOXX and VIX

The main state dependent variable is VSTOXX, and I use it as an uncertainty proxy measure. VSTOXX is an option implied volatility index based on the EURO STOXX 50 index options. It is designed to indicate European equity market volatility over the near future. I obtained the VSTOXX data from STOXX historical data.

In the cleaned dataset, the variables include the raw VSTOXX value and a standardised version, VSTOXX_z. The VSTOXX data were merged to the ECB event dataset by date and for each ECB event, the matched VSTOXX value is the previous available market close before the event date. This was done so that the uncertainty variable reflects the market state before the ECB announcement rather than the market reaction after the announcement. The cleaned dataset also records the source date of the matched VSTOXX observation and the number of days between the VSTOXX observation and the ECB event date.

The robustness uncertainty variable is VIX, an option-implied index based on S&P 500 index options which is available from the FRED database. As with VSTOXX, each ECB event was matched to the previous available VIX close before the event date. The cleaned VIX variable was then merged into the press conference wide and long datasets. Both VSTOXX and VIX are used as pre-event state variables. The cleaned datasets include standardised versions of the volatility variables so that interaction coefficients can be interpreted in relation to a one-standard-deviation change in the uncertainty measure.

The standardised value is calculated as Z = (X-mean of X) / standard deviation of X, where X is the raw volatility index. This means that the standardised versions measure how many standard deviations the volatility index is above or below its sample mean. The standardisation is implemented before the country event panel so that the same ECB event panel is not given any additional weight since all four countries are included in the panel.

3.4 Fiscal Vulnerability Variable: Debt-to-GDP

The fiscal vulnerability variable is government debt-to-GDP. I do not treat this variable as another market uncertainty measure. It is a country level fiscal variable that captures the public-debt burden relative to national output. The debt-to-GDP data were taken from Eurostat government debt data. The analysis keeps the four countries used in the sovereign yield analysis: Germany, France, Italy and Spain. A standardised version of debt-to-GDP is also created for the regression analysis.

3.5 Dataset Construction and Sample

The cleaned data are organised in two main formats. The first is a wide event-level dataset. In this format, each row corresponds to one ECB event, and the yield responses for Germany, France, Italy and Spain appear as separate variables. This structure is used for the individual country level regressions. The second format is a long country-event dataset. In this structure, each row corresponds to one country on one ECB event date. The long dataset includes the country code, country name, periphery indicator, the 10-year yield response, and the relevant ECB surprise variables. This format is used for pooled panel regressions, core/periphery specifications, period splits, and the debt-to-GDP extension. The press release and press conference are not separate policy events but rather two communication windows within the same ECB policy event. The press-release dataset contains 242 ECB events from 2002 to 2025. The press-conference dataset contains 237 ECB events from 2002 to 2025. With four countries, the press-conference long dataset contains 948 country-event observations before dropping observations due to missing matched state variables. The VSTOXX-matched press conference sample contains 232 ECB events, corresponding to 928 country-event observations in the long format. The number of observations therefore varies slightly across specifications depending on whether the model uses the full EA-MPD event sample or requires matched VSTOXX, VIX or debt-to-GDP data. For period-split analysis, the press conference panel is divided into three periods: 2002–2009, 2010–2018 and 2019–2025. These period indicators are created from the ECB event date. The final event date in the sample is 30 October 2025 and the dataset does not include standard low-frequency macroeconomic control variables such as GDP growth, inflation or unemployment. This is because the analysis is based on narrow event window changes around ECB announcements. The purpose of the dataset is to isolate the immediate market reaction to unexpected ECB news rather than to explain low-frequency sovereign yield movements using broader macroeconomic fundamentals.

3.6 Data Cleaning and Management

The data cleaning process involved five main steps. First, the EA-MPD event-level files were organised into press release and press conference datasets. Second, the sovereign yield response variables were kept for Germany, France, Italy and Spain across the relevant maturities. Third, the VSTOXX and VIX daily volatility series were cleaned, date formatted and matched to ECB events using the previous available close. Fourth, the Eurostat debt-to-GDP data were imported, reshaped, cleaned and merged to the country-event panel using the previous quarter. Fifth, final wide and long datasets were saved for the country level and pooled panel specifications. The cleaned datasets therefore contain the ECB event date, the relevant monetary policy surprise variables, the sovereign yield response variables, the pre-event uncertainty variables, and country identifiers. In the long dataset, the panel unit is the country event pair.

3.7 Reliability and Limitations of the Data

The monetary policy surprise and financial response variables are based on high frequency event window data. This improves the reliability of the identification because the measured financial market responses are tied closely to ECB policy events. The use of market-based surprise factors also avoids relying on subjective manual coding of ECB communication. There are still limitations. VSTOXX and VIX are daily variables, while the EA-MPD responses are measured at higher frequency around ECB events. Matching the previous available close reduces announcement contamination, but it does not make high uncertainty and low uncertainty events identical in every other respect. Debt-to-GDP is lower frequency quarterly fiscal data, so it changes more slowly than market volatility measures. I therefore interpret it as a fiscal vulnerability variable rather than as a high frequency market state variable. Finally, missing values reduce the matched uncertainty sample slightly. The main press conference sample contains 237 events, while the VSTOXX-matched sample contains 232 events. I report this difference because the number of observations varies across specifications depending on the availability of the relevant state variable.

4 Empirical Methodology

In this section, I explain the empirical strategy I use to estimate the response of European sovereign yields to ECB monetary policy surprises. This section builds directly on the data described in the previous section and separates the empirical design from the reporting and interpretation of results. As mentioned previously, the main idea is to relate changes in sovereign yields around ECB event windows to monetary policy surprise factors and then test whether these responses vary with changes in uncertainty or fiscal vulnerability.

4.1 Estimation Approach

For all regressions, I use Ordinary Least Squares (OLS). In the context of this dissertation, OLS is appropriate because the empirical objective is to estimate the linear event window response of sovereign yields to measured ECB monetary policy surprises. The dependent variable is not the level of sovereign yields, but high frequency change in yields around ECB policy events while the explanatory variables are monetary policy surprise factors measured over the same event structure. The OLS coefficient can therefore be interpreted to reflect average change in sovereign yields associated with a one-unit unexpected ECB surprise within the event window. The identifying strength of my approach comes from this specific high frequency event study design, which limits the slower macroeconomic developments, for example controls such as inflation, unemployment or GDP growth, instead of imposing a complex structural model.

For country level regressions, I use heteroskedasticity robust standard errors and for the pooled country event panel regression, standard errors are clustered around the ECB event date since all four countries are exposed to the same ECB shock on a given policy day. Such a clustering choice allows for correlation in the regression errors across countries within the same ECB event.

4.2 Baseline Country Level Specification

Empirical Objective

The baseline empirical model is:

$$\Delta y_{i,t} = \alpha + \beta S_t + \varepsilon_{i,t}$$

Here, $\Delta y_{i,t}$ is the high-frequency change in sovereign yield of country i around ECB event t. S_t represents an ECB monetary policy surprise and β is the average response of sovereign yields to the relevant policy surprise.

In the press release check S_t represents the target surprise while in the main press conference analysis, the specification incorporates both timing and forward guidance surprises together:

$$\Delta y_{i,t} = \alpha + \beta_1 \text{Timing}_t + \beta_2 \text{FG}_t + \varepsilon_{i,t}$$

The coefficient β_1 measures the average sovereign yield response to a timing surprise, while β_2 measures the average response to a forward guidance surprise. I estimate this baseline model first to show whether ECB press conference communication moves yields in the expected direction before introducing any state-dependent interactions.

I estimate the baseline regressions separately for Germany, France, Italy and Spain. This country level approach is appropriate because the sovereign markets differ in liquidity and fiscal risk and so estimating the countries separately avoids assuming at the onset that all the countries respond similarly to identical ECB surprises.

4.3 Economic Interpretation of Sovereign Yield Responses

Sovereign yields can be defined as the market return that investors require for lending to a government over a particular maturity. For example, in the context of Italy, the 10-year sovereign yield is the return investors require to hold Italian government debt for ten years.

The inverse relationship between bond yields and bond prices is crucial. A bond comprises fixed future payments commonly called a coupon payment. If the bond price rises, this fixed payment represents a relatively lower return now. If the price falls, the same fixed payments indicate a higher return.

ECB monetary policy directly affects sovereign yields because it leads to change in future expectations about short-term interest rates, inflation, and term premia. If the ECB policy surprise during the press conference is hawkish, markets might change/update their expectations of future policy rates upward. Therefore, existing bonds with fixed coupon payments become less attractive, and bond prices fall while yields rise.

The basic transmission chain is therefore:

Hawkish ECB surprise $\rightarrow$ higher expected future rates $\rightarrow$ lower bond prices $\rightarrow$ higher sovereign yields

The dissertation examines whether this chain works equally under all market conditions. The state-dependence question is regarding whether the same ECB surprise has the same impact when uncertainty or financial stress is higher.

4.4 State-dependent specification

The state-dependent extension to the baseline model is:

$$\Delta y_{i,t} = \alpha + \beta S_t + \gamma Z_t + \delta (S_t \times Z_t) + \varepsilon_{i,t}$$

Here, Z_t is a state variable, such as VSTOXX or VIX. The key coefficient is δ, testing whether the effect of an ECB surprise changes due to higher levels of uncertainty or global market volatility. A positive interaction indicates that the sovereign yield response to the ECB surprise would be stronger when the state variable is higher. A negative interaction would alternatively indicate that the yield response is weaker when the state variable is higher.

In the principal analysis, Z_t is VSTOXX, a proxy measure for market uncertainty. In the robustness analysis, Z_t is replaced by VIX. In an additional extension, the state variable is replaced by country level debt-to-GDP, which captures fiscal vulnerability which is also a source of fiscal uncertainty.

4.5 Why Amplification and Dampening Are Both Plausible

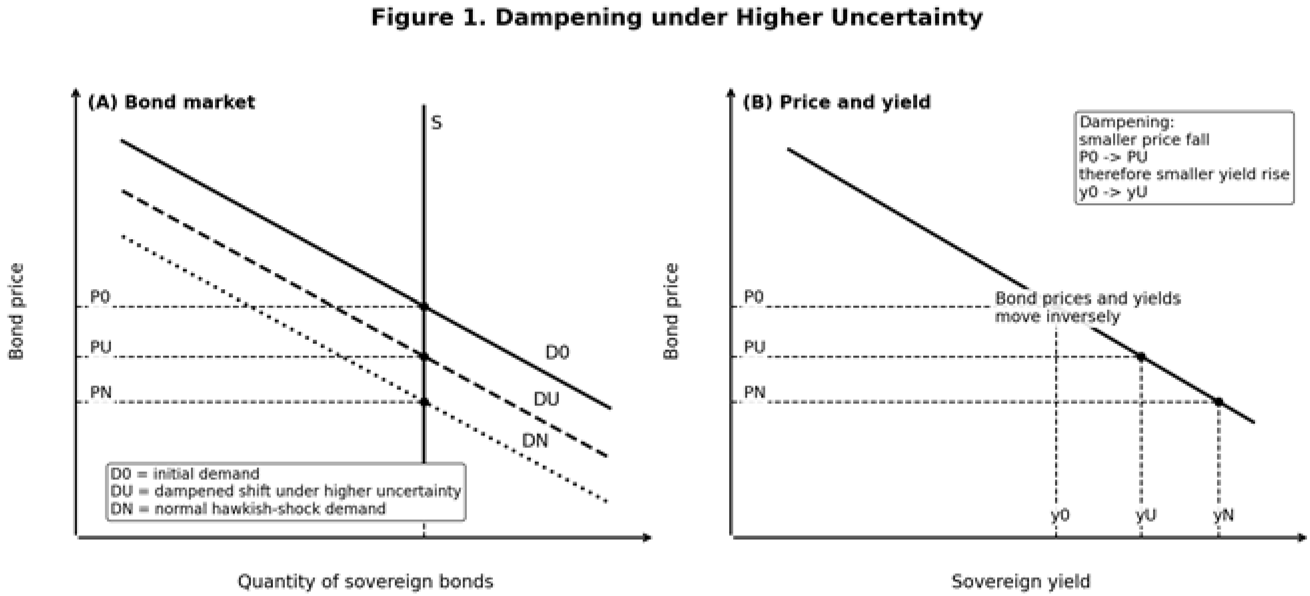
Theoretically, the impact of uncertainty on monetary policy transmission is not unidirectional. Both amplification and dampening are economically plausible. Therefore, the sign of the interaction term must be empirically tested and economically interpreted.

4.5.1 Dampening Channel

The dampening channel works through a separate mechanism. In periods of high uncertainty or stress, the ECB signal is not necessarily priced in isolation. Markets may simultaneously price recession risk, liquidity conditions, safe-haven flows and the possibility that a hawkish ECB policy may later be reversed if the economic outlook deteriorates. These forces do not just add random noise, but several of them work in a specific direction that offsets the usual yield-increasing result of a hawkish communication.

First, core sovereign yields can be mechanically compressed by flight-to-quality demand. When uncertainty is high, investors may often prefer safer and more liquid bonds, like German Bunds. This demand raises bond prices and lowers yields, thereby partly offsetting the effect of a hawkish ECB surprise. Second, market depth may deteriorate during stressed periods. Weaker market depth and wider bid-ask prices may cause prices to adjust more slowly within the narrow event window, thereby muting the measured yield response. Third, hawkish guidance could also be viewed as more conditional in an uncertain environment. This means investors may think that if growth weakens and financial stress intensifies, the ECB will not be able to maintain a tighter path. That lowers the perceived persistence of the ECB policy signal.

Figure 1. Dampening under higher uncertainty



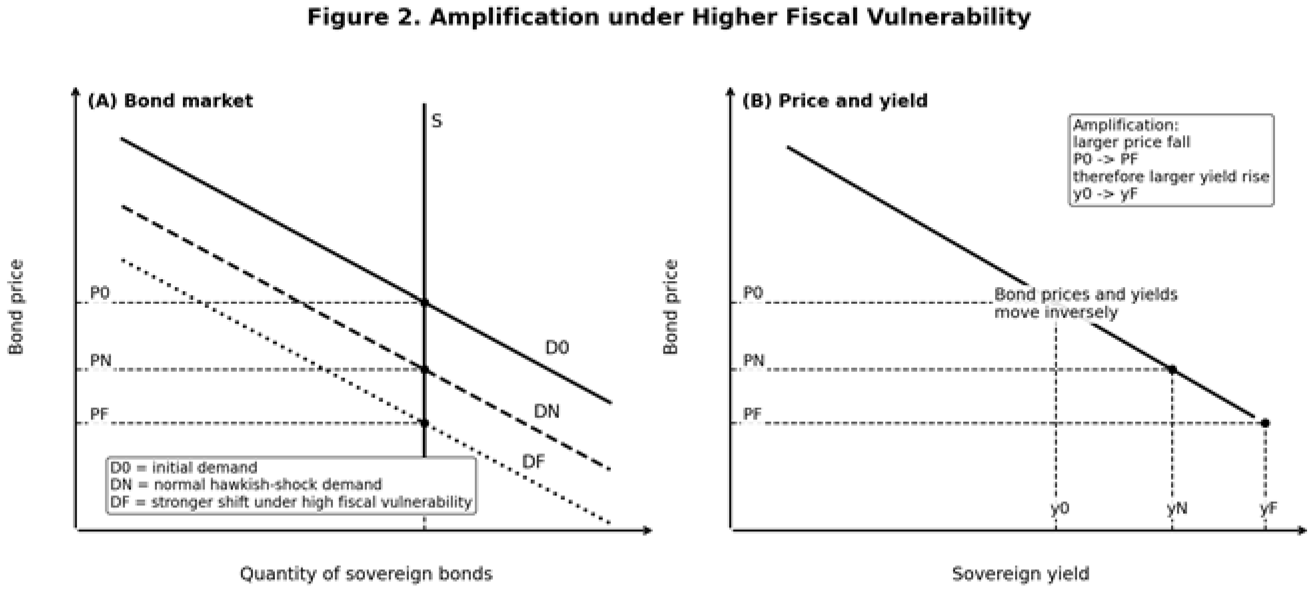
Explanation. Panel A shows the sovereign bond market. The initial equilibrium is given by the intersection of fixed bond supply and the initial demand curve, which produces the initial bond price, P0. A hawkish ECB surprise reduces demand for existing fixed coupon bonds because markets revise expected future interest rates upward. Under normal conditions, this demand shift is relatively strong and the equilibrium price falls from P0 to PN. However, under higher uncertainty, the same hawkish shock is partly offset by other forces, such as safe-haven demand, recession concerns, lower inflation expectations, liquidity effects, or expectations that future tightening may later be reversed. As a result, bond demand falls by less, so the equilibrium price falls only from P0 to PU rather than all the way to PN.

Yield interpretation. Because the price fall is smaller under higher uncertainty, the yield rise is also smaller.

4.5.2 Amplification Channel

Amplification works through the channels of risk premia and fiscal vulnerability. If stress or uncertainty is high in the economy, investors may become nervous about risky sovereign debt. A hawkish ECB surprise can raise expected future interest rates and increase concerns about debt servicing costs, refinancing risk and fiscal vulnerability. This is of relevance in the context of higher debt or higher spread sovereigns. In such cases, investors are prone to requiring higher risk premia. As a result, bond prices fall more sharply, and yields rise more strongly compared to the general case.

Figure 2. Amplification under higher fiscal vulnerability



Explanation. Panel A again starts from the initial bond market equilibrium at price P0. A hawkish ECB surprise lowers bond demand under normal conditions, reducing the price to PN. When fiscal vulnerability is higher, however, the same policy signal can trigger a larger reassessment of refinancing costs, debt sustainability, and risk premia. As a result, investors become more sensitive to the expected future path of interest rates. Therefore, bond demand falls further and the equilibrium price drops from P0 to PF, which is lower than the normal shock price PN.

Yield interpretation. Panel B demonstrates the corresponding inverse price-yield relationship. Because the price fall is larger under higher fiscal vulnerability, the yield rise is also larger.

The important point is that the two channels diverge after the initial rate expectation stage. When higher interest rates increase sovereign risk premia, amplification occurs while dampening occurs when liquidity conditions, safe-haven demand, recession concerns, and policy-reversal possibilities offset part of the yield response. The VSTOXX and VIX results indicate that in conditions of high uncertainty and systemic stress, the dampening channel dominates. The debt-to-GDP extension, however, points to the fact that fiscal vulnerability can produce amplification for forward guidance shocks specifically.

4.6 Pooled Country Event Panel Specification

I organise the empirical strategy in two ways. First, I estimate individual country level regressions for Germany, Italy, France and Spain. This allows me to examine the response of each sovereign market separately. Second, I estimate the pooled country level panel which stacks the four countries together and includes country fixed effects. The primary objective here is to test whether the main transmission result survives when all four countries are treated as one country-event dataset. The panel specification is useful to analyse and summarise the common response across all four countries while allowing each one to have its own average response level. Standard errors remain clustered by ECB event date since all countries are exposed to the same ECB shock on a given policy day.

The baseline panel model is:

$$\Delta y_{i,t} = \alpha_i + \beta_1 \text{Timing}_t + \beta_2 \text{FG}_t + \varepsilon_{i,t}$$

where α_i represents country fixed effects.

In the state-dependent panel specification, VSTOXX or VIX is added together with interaction terms between the state variable and timing and forward guidance:

$$\begin{aligned}\Delta y_{i,t} = {} & \alpha_i + \beta_1 \text{Timing}_t + \beta_2 \text{FG}_t + \gamma Z_t \\ & + \delta_1 (\text{Timing}_t \times Z_t) + \delta_2 (\text{FG}_t \times Z_t) + \varepsilon_{i,t}\end{aligned}$$

The pooled panel summarises the common response across countries while controlling for permanent average differences between countries through fixed effects.

4.7 VIX Robustness Specification

I use VIX as the main robustness check. Instead of euro area equity market volatility, VIX captures global market volatility. Therefore, it is used to test whether the dampening result is tied only to VSTOXX, or whether it remains consistent when a different uncertainty proxy measuring international uncertainty is used as the state variable.

The interaction structure I use for the VIX robustness specification is the same as the VSTOXX structure, the only change being that Z is replaced by standardised VIX. This ensures consistent interpretation of the interaction coefficients across the baseline uncertainty measure and the broader international financial volatility proxy.

4.8 Term Structure Extension

The objective of the term structure extension is to examine whether the state-dependent result is consistent across different sovereign yield maturities. This is useful because different maturities help capture different parts of monetary policy transmission. For example, shorter maturities are considered more closely related to short-term ECB rates, while longer maturities may capture long-run inflation expectations, term premia, liquidity premia and sovereign risk-premia.

The specification is modified by replacing the dependent variable with the 2-year and 5-year sovereign yield responses respectively. This allows the analysis to compare whether surprises affect the short, medium and long term parts of the sovereign yield curve differently.

4.9 Core/Periphery and Spread Specifications

The core vs periphery analysis compares Germany and France with Italy and Spain. Germany and France are considered as core/lower-risk sovereigns, while Italy and Spain are treated as peripheral/higher-spread sovereign nations. Although the individual country regressions remain important, this classification is useful because the four countries under consideration do not represent identical types of sovereign risk.

In the panel specification, a periphery indicator is interacted with the ECB surprise variables to test whether Italy and Spain indicate a different average response compared to Germany and France. This enables the analysis to distinguish common euro area interest rate expectation channel, and the additional periphery specific component. I also use spread specifications as supporting checks, to check if ECB communication changes the responses between peripheral and core sovereign yields.

4.10 Period Split Specification

To determine whether the baseline results vary across the broad monetary policy regimes, I use a period-split panel. I split the country event panel into 2002-2009, 2010-2018 and 2019-2025. The division separates the early euro area/pre crisis and Global Financial Crisis period, the euro area crisis and unconventional period, and the post-2019 period covering the COVID-19 pandemic, inflation shock, and ECB tightening cycle.

The same baseline panel model is estimated separately within each period, making it feasible to compare whether the response to timing and forward guidance surprises is stable across regimes or concentrated in specific parts of the sample. I interpret the period split estimates with reservations because splitting the sample reduces the number of ECB events available in each regression, thereby reducing the power of the regressions.

4.11 Debt to GDP Fiscal Vulnerability Specification

For the debt-to-GDP extension, I use a different state variable than VSTOXX. Debt-to-GDP is not a high-frequency market volatility measure. Instead, it is a measure of fiscal vulnerability which captures uncertainty regarding debt sustainability and future financing costs. The primary objective is to examine whether the amplification channel becomes the dominant channel when the chosen state variable is closer to sovereign debt exposure.

The debt-to-GDP section makes use of the pooled panel structure while including country fixed effects. This is essential because both Italy and Spain have persistently higher debt ratios than

Germany and France. Country fixed effects control for permanent average differences across countries and so the interaction terms can be interpreted as differences in the slope of the yield response with respect to fiscal vulnerability, and not simply as level differences between countries.

The interaction model can be written as:

$$\begin{aligned}\Delta y_{i,t} &= \alpha_i + \beta_1 \text{Timing}_t + \beta_2 \text{FG}_t + \gamma \text{Debt}_{i,t} \\ &\quad + \delta_1 (\text{Timing}_t \times \text{Debt}_{i,t}) + \delta_2 (\text{FG}_t \times \text{Debt}_{i,t}) + \varepsilon_{i,t}\end{aligned}$$

Here, $\text{Debt}_{i,t}$ is the standardised debt-to-GDP ratio for country i before ECB event t. A positive interaction coefficient here indicates that fiscal vulnerability strengthens the sovereign yield response to the relevant ECB surprise. This specification is different from the VSTOXX and VIX models since it tests fiscal exposure instead of broad market uncertainty.

4.12 Equity Market Checks

Even though equity market checks are not central to this analysis, I include them as additional financial market evidence. To perform these checks, I replaced the sovereign yield response with the relevant equity market response while maintaining the same logic. The purpose is to observe whether financial market variables show related responses to ECB surprises, while keeping sovereign yields as the dependent variable.

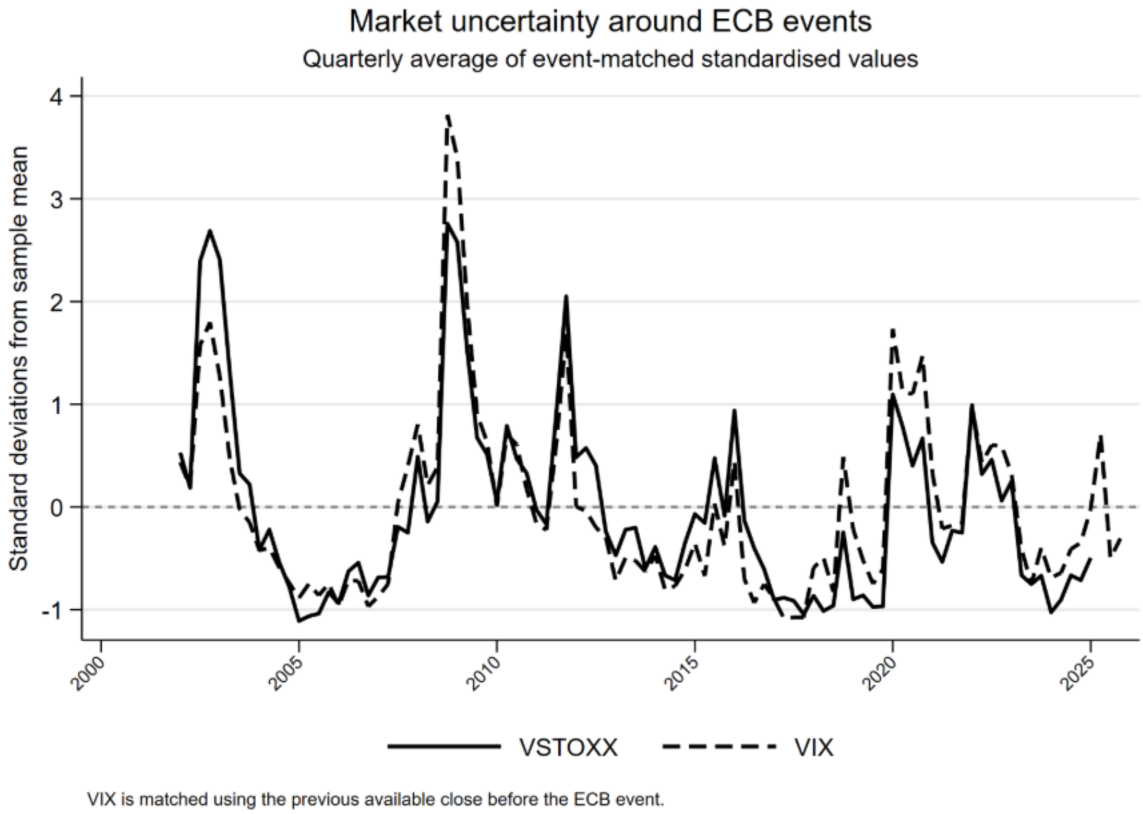
Overall, the methodology uses the same event-study logic across the main analysis and corresponding extensions. I use the baseline regressions to estimate the average response to ECB surprises and the interaction specifications to test whether that response changes across states, and the panel, term structure, period split and fiscal vulnerability to examine whether the central result is robust across countries, maturities, periods and state variables.

5 Results and Interpretation

This section presents the empirical results on the transmission of ECB monetary policy surprises to euro-area sovereign yields. The results are reported and organised in stages. I begin by reporting the baseline press conference estimates, followed by an examination of whether the response varies with market uncertainty using VSTOXX. I then report pooled country event level estimates, VIX robustness results, term-structure extensions, core/periphery specifications, period split estimates, the debt-GDP extension, and finally the equity market checks. The primary object of interest is the event window change in sovereign yields. Therefore, the results are examined and interpreted as evidence on immediate financial market pricing around ECB communication windows instead of long run yield regressions.

Figure 5.1 shows the uncertainty environment surrounding the ECB events. The figure provides context regarding the state variables used in the corresponding interaction regressions.

Figure 5.1. VSTOXX and VIX around ECB events



quarterly averages of event-matched standardised values. VIX is matched using the previous available close before each ECB event.

5.1 Baseline Press Conference Results

I use the baseline regressions to estimate the average response of sovereign yields to ECB monetary policy surprises without state interactions. This is the natural starting point since it helps demonstrate whether the timing and forward-guidance factors move sovereign yields in the expected direction before any state-dependent interactions are introduced. Table 5.1 presents these baseline press conference estimates for the four countries.

Table 5.1. Baseline press conference regressions: 10Y sovereign yield responses

Variable	Germany	France	Italy	Spain
Timing surprise	0.390***	0.342***	0.291**	0.282***
SE	(0.0759)	(0.0797)	(0.131)	(0.103)
Forward-guidance surprise	0.495***	0.501***	0.496***	0.471***
SE	(0.0574)	(0.0669)	(0.0863)	(0.0715)
Observations	237	237	237	237
R-squared	0.375	0.344	0.126	0.172

*Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

The country level results indicate that both timing and forward guidance surprises have positive and significant effects on the 10-year sovereign yields across Germany, France, Italy and Spain. The respective timing effects range between 0.282 and 0.390, while the forward guidance effects are observed to be tightly clustered around 0.471 and 0.501. The baseline results therefore point toward a broad European sovereign yield response to ECB press conference communication. Since forward guidance effects are especially similar across countries, this suggests that unexpected information regarding future policy path may be priced across sovereign bond markets as a common ECB communication shock.

The positive baseline coefficients are therefore consistent with the standard bond pricing channel through which hawkish monetary policy communication increases sovereign yields.

In the context of the press release, the target surprise is weaker in the broader set of results, which can be interpreted as immediate policy decision is often more anticipated by the market. By contrast, the press conference includes information regarding timing, persistence, and future policy direction and is therefore central to the analysis. The baseline results support the focus on timing and forward guidance shocks as the main press conference channels in the rest of the empirical analysis.

5.2 State Dependence with VSTOXX

For the main state dependent specification, I interact timing and forward guidance surprises with VSTOXX. These interaction terms answer the central empirical question: does the response of sovereign yields to ECB communication become stronger or weaker when uncertainty is higher? Table 5.2 reports the full country by country VSTOXX interaction specification, therein the main coefficients of interest are Timing × VSTOXX and Forward Guidance × VSTOXX.

Table 5.2. Country level VSTOXX interaction regressions: 10Y sovereign yield responses

Variable	Germany	France	Italy	Spain
Timing surprise	0.447***	0.427***	0.430***	0.393***
SE	(0.0828)	(0.0877)	(0.129)	(0.0989)
Forward-guidance surprise	0.529***	0.529***	0.515***	0.489***
SE	(0.0507)	(0.0591)	(0.0779)	(0.0659)
VSTOXX, standardised	-0.0457	0.165	0.725	0.494**
SE	(0.140)	(0.178)	(0.475)	(0.230)
Timing x VSTOXX	-0.124**	-0.188***	-0.311**	-0.245***
SE	(0.0620)	(0.0666)	(0.148)	(0.0939)
Forward guidance x VSTOXX	-0.211***	-0.188***	-0.146	-0.133**
SE	(0.0493)	(0.0594)	(0.103)	(0.0652)
Observations	232	232	232	232
R-squared	0.412	0.375	0.153	0.194

Negative interaction coefficients indicate that higher VSTOXX dampens the yield response to ECB communication surprises.

A clear dampening pattern is observable here. The timing × VSTOXX coefficients are statistically significant and negative for all four countries, ranging from -0.124 for Germany to -0.311 for Italy. The forward-guidance × VSTOXX coefficients for all countries are also negative. However, they are statistically significant for three out of the four countries, Italy being the exception. The differences in magnitudes likely show differences in liquidity, sensitivity to sovereign risk and how strongly market uncertainty affects each country's bond market.

The interpretation should not be that higher uncertainty tends to mechanically lower sovereign yields. The dependent variable is the event window yield change, and therefore the interaction term measures how the yield change response to an ECB surprise changes when VSTOXX is higher. Since the baseline effect of timing and forward guidance is positive, this means that the yields still rise but not to the extent as would be the case if uncertainty was not high. The same logic is symmetrically applicable for a dovish surprise as well. Under higher uncertainty a dovish surprise would be expected to lower yields by less in absolute magnitude. Just to reiterate the effect, the interaction is a statement regarding strength of transmission and not the unconditional direction of yields.

Economically, this dampening result is plausible because in high uncertainty periods, the ECB signal is not priced in isolation. Four major factors could be attributed to the cause of this dampening result. First, the safe-haven channel which applies most directly to Germany, due to which demand can compress German Bund yields, second, liquidity may weaken and bid-ask spreads may widen, making immediate price discovery less complete, and third, in periods of uncertainty, markets may consider that a hawkish ECB policy may later be reversed because uncertainty may raise the probability of weaker growth and negatively impact the economy. Additionally, as observed with the recent inflation shock, uncertainty may coincide with periods of high inflation. In this case, a hawkish policy surprise may lower inflation expectations, especially at the longer maturities. These mechanisms can therefore offset part of the usual yield increasing effect of a hawkish surprise.

It is useful to highlight certain country level results as well. Italy is one of the countries where intuitive reasoning could imply amplification because of Italy's larger debt burden and consequently higher sovereign-risk. However, VSTOXX captures broad euro area market uncertainty and not Italy specific fiscal vulnerability. Italy's timing interaction is negative and statistically significant, while the forward guidance interaction is also negative but less precisely estimated. It can be inferred that broad uncertainty dampens the timing response of Italy, while there is no significant effect on forward guidance. The Italian result is useful because it highlights the choice importance of the state variable i.e. a broad uncertainty measure weakens the immediate impact of the ECB communication even in a country where amplification is theoretically plausible. The debt-to-GDP extension discussed later involves the direct examination of fiscal vulnerability or fiscal uncertainty as a state variable. Debt-to-GDP can be interpreted like this since a higher debt burden may increase uncertainty about debt sustainability, refinancing pressure and sensitivity of sovereign yields to persistent future rate guidance.

Spain also provides certain interesting results. Its timing and forward guidance interactions with VSTOXX are both negative and statistically significant, thereby supporting the dampening interpretation. At the same time, the VSTOXX coefficient (not the interaction coefficient) in the Spanish regression is positive and significant i.e. when the measured timing and forward guidance surprises are zero, high uncertainty can be associated with higher Spanish sovereign yield movements. This is consistent with some sovereign risk pressure in high uncertainty ECB event environments. The negative interaction terms however show that the marginal response of Spanish sovereign yields to the measured ECB surprise is still weaker when VSTOXX is high. As a result, it is observed that Spain shows both an uncertainty-related upward pressure and dampened transmission of the communication shock itself. However, the main dissertation claim is dependent on the interaction terms and not on the standalone VSTOXX coefficient.

5.3 Pooled Country Panel Results

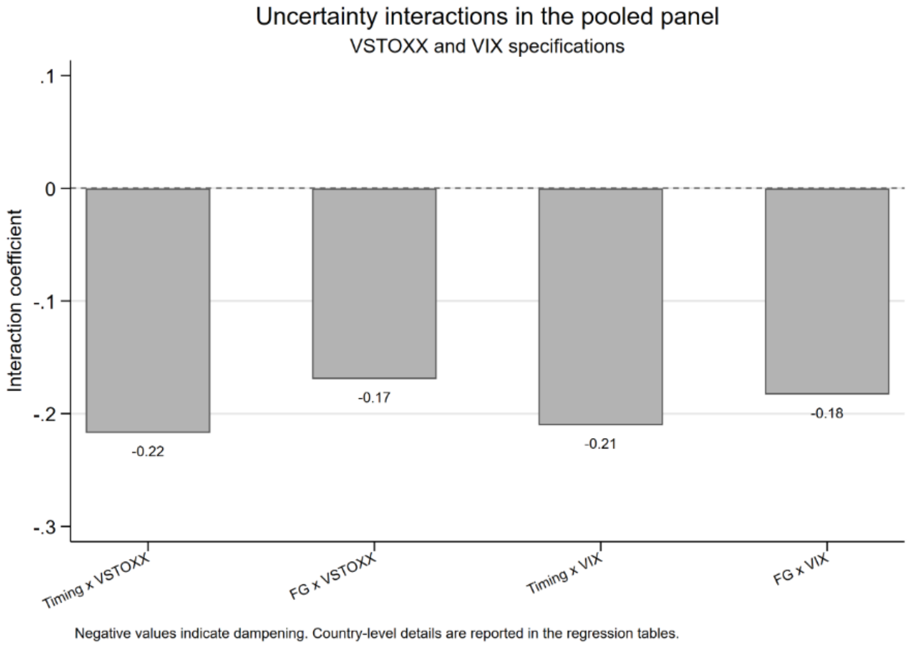
Table 5.3 reports pooled country event panel results across the baseline, VSTOXX, and VIX specifications. Figure 5.2 demonstrates the uncertainty interaction coefficients so that the dampening pattern is observed visually along with the full regression evidence.

Table 5.3. Pooled country-event panel: baseline, VSTOXX and VIX

Variable	Baseline	VSTOXX	VIX
Timing surprise	0.326***	0.424***	0.427***
SE	(0.0883)	(0.0921)	(0.0967)
Forward-guidance surprise	0.491***	0.516***	0.556***
SE	(0.0654)	(0.0578)	(0.0549)
Uncertainty index		0.335	0.265
SE		(0.213)	(0.226)
Timing x uncertainty		-0.217***	-0.210***
SE		(0.0775)	(0.0716)
Forward guidance x uncertainty		-0.169***	-0.183***
SE		(0.0591)	(0.0590)
Observations	948	928	948
R-squared	0.211	0.231	0.242

Country fixed effects are included; standard errors are clustered by ECB event date.

Figure 5.2. VSTOXX and VIX interaction effects in the pooled panel



Negative interaction coefficients support the dampening interpretation under market uncertainty.

The panel baseline confirms the country level findings that ECB press conference communication raises sovereign yields. Timing and forward guidance surprises are both positive and significant in this specification. The timing coefficient is 0.326 while the forward-guidance coefficient is 0.491, with 948 country-event observations. The pooled coefficient does not replace the country level results, but it summarises the common component of the response across the four countries.

This specification also strongly supports the state-dependent results. During periods of uncertainty, timing and forward guidance surprises continue to be positive and significant. The interaction terms with VSTOXX are negative and significant. The coefficients for timing × VSTOXX and forward-guidance × VSTOXX are –0.217 and –0.169 respectively. The panel result is useful because it demonstrates that the dampening result is not a simple country-by-country pattern.

I also estimate a panel specification allowing for core/periphery differences within the pooled framework. The periphery interactions are not statistically significant, meaning that Italy and Spain do not display a consistently different average response from Germany and France once the common ECB surprise effect and country fixed effects are included. When VSTOXX and core/periphery interactions are included together, the VSTOXX interactions remain negative and significant, while the periphery interactions remain insignificant. This indicates that the main state-dependence result is the broad dampening effect of uncertainty, rather than a simple story in which the periphery always reacts differently from the core.

5.4 Robustness Using VIX

Table 5.4. Country-level VIX robustness regressions: 10Y sovereign-yield responses

Variable	Germany	France	Italy	Spain
Timing surprise	0.448***	0.430***	0.438***	0.393***
SE	(0.0816)	(0.0886)	(0.140)	(0.108)
Forward-guidance surprise	0.571***	0.573***	0.555***	0.526***
SE	(0.0453)	(0.0513)	(0.0797)	(0.0694)
VIX, standardised	-0.0817	0.107	0.699	0.334
SE	(0.126)	(0.182)	(0.526)	(0.220)
Timing x VIX	-0.136***	-0.190***	-0.287**	-0.227***
SE	(0.0495)	(0.0618)	(0.132)	(0.0823)
Forward guidance x VIX	-0.217***	-0.206***	-0.158	-0.152**
SE	(0.0407)	(0.0553)	(0.106)	(0.0667)
Observations	237	237	237	237
R-squared	0.428	0.393	0.162	0.199

VIX is used as a global financial-market uncertainty robustness check. The pattern is very close to the VSTOXX result.

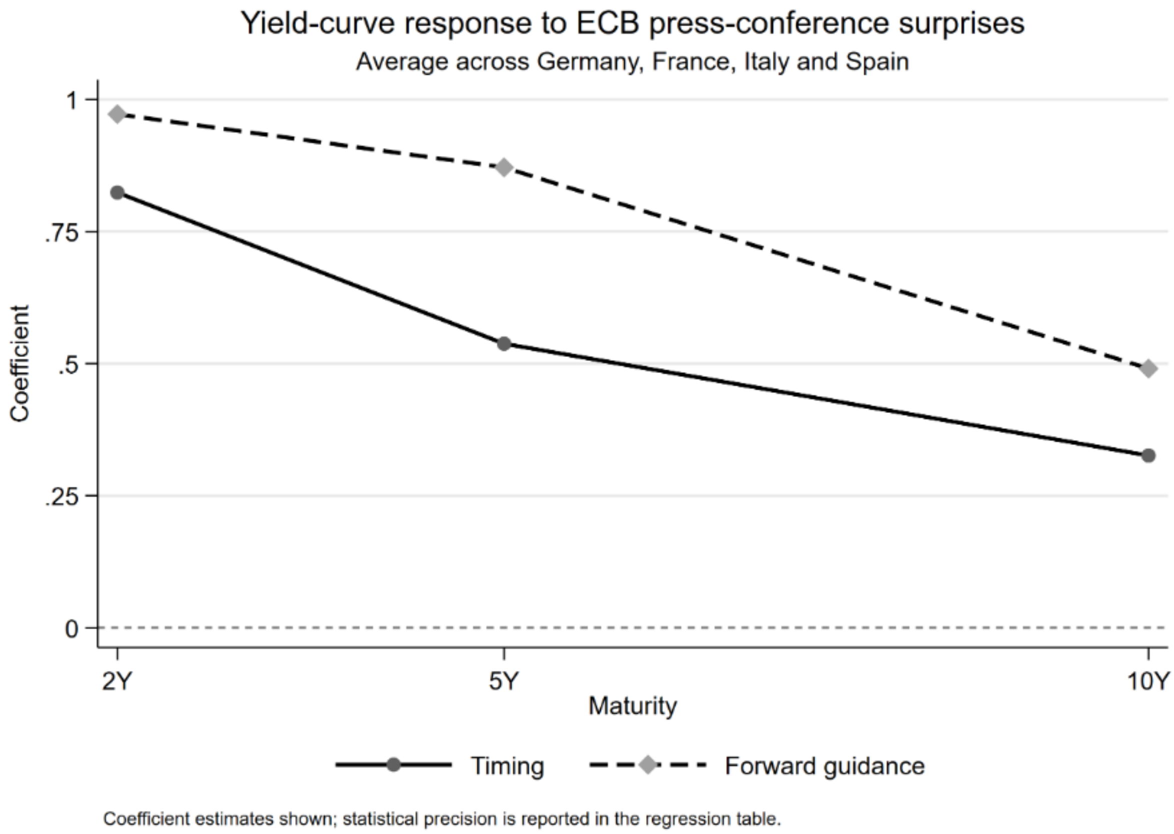
The country level VIX results are in line with the VSTOXX findings. The timing × VIX interaction is negative and statistically significant for Germany, France, Italy and Spain, with coefficients of –0.136, -0.190, -0.287 and -0.227 respectively. The forward-guidance × VIX interaction is also negative and statistically significant for all four countries except Italy. It is negative but not statistically significant for Italy. The pattern is therefore very close to the VSTOXX specification. The interpretation is also similar. The VIX results reinforce the view that market uncertainty dampens the transmission of ECB communication shocks to sovereign yields.

The VIX results should be interpreted as a supporting robustness check rather than as a fully independent test because global and European volatility tend to move together during major periods of stress. The point of the VIX robustness check is therefore not that it is uncorrelated with VSTOXX, but that it helps to capture broader international financial volatility.

5.5 Term Structure Results

Figure 5.3 and Table 5.5 report on the term structure results across different yield maturities to show whether the baseline communication effect and the uncertainty interaction are concentrated at one maturity or appear across the yield curve.

Figure 5.3. Yield-curve response across maturities



Shorter maturities respond more strongly to ECB press-conference surprises than 10-year yields.

Table 5.5. Term structure with VSTOXX interactions

Variable	Average 2Y	Average 5Y	Average 10Y
Timing surprise	0.893***	0.639***	0.424***
SE	(0.0717)	(0.0914)	(0.0927)
Forward-guidance surprise	0.990***	0.905***	0.516***
SE	(0.0362)	(0.0367)	(0.0582)
VSTOXX, standardised	0.155	0.288	0.335
SE	(0.111)	(0.212)	(0.215)
Timing x VSTOXX	-0.147***	-0.215**	-0.217***
SE	(0.0496)	(0.0862)	(0.0781)
Forward guidance x VSTOXX	-0.117***	-0.199***	-0.169***
SE	(0.0395)	(0.0636)	(0.0595)
Observations	232	232	232
R-squared	0.814	0.642	0.311

The response is strongest at the short end of the yield curve, but negative VSTOXX interactions appear across maturities.

The results indicate that both timing and forward guidance surprises affect the 2-year and 5-year sovereign yields. The coefficients are larger at the shorter end of the curve, timing effects decline from 0.893 for the average 2-year yield to 0.424 for the average 10-year yield, while forward-guidance effects decline from 0.990 to 0.516. The results are economically intuitive. The 2-year yield is most directly connected to the expected ECB policy over the near future, and so it reacts more strongly to information about the timing and future path of policy.

The VSTOXX interactions remain negative across the term structure implying that dampening is not only a feature of the 10-year benchmark. Higher uncertainty weakens the response at both shorter and longer maturities. The R-squared values also fit the economic interpretation. The model explains a much larger share of 2-year average yield response (R^2 = 0.814) than the average 10-year response (R^2 = 0.311) because long maturity yields contain additional components that are not fully captured by the ECB surprise factors alone.

5.6 Core vs Periphery and Spread Results

Table 5.6 Core/Periphery 10Y responses with VSTOXX interactions

Variable	Core 10Y	Periphery 10Y	Periphery - Core
Timing surprise	0.437***	0.412***	-0.0254
SE	(0.0837)	(0.113)	(0.0710)
Forward-guidance surprise	0.529***	0.502***	-0.0275
SE	(0.0543)	(0.0708)	(0.0489)
VSTOXX, standardised	0.0596	0.610*	0.550*
SE	(0.144)	(0.336)	(0.287)
Timing x VSTOXX	-0.156***	-0.278**	-0.122
SE	(0.0600)	(0.116)	(0.0979)
Forward guidance x VSTOXX	-0.199***	-0.139*	0.0601
SE	(0.0525)	(0.0800)	(0.0643)
Observations	232	232	232
R-squared	0.405	0.178	0.0306

Core and periphery both respond to ECB communication. The additional periphery-core spread effect is weaker than the common yield response.

The ECB communication channel affects both groups as both core and periphery yields respond positively to press conference timing and forward guidance shocks. The VSTOXX interactions are negative for both core and periphery yields indicating that the dampening result is not restricted to any one group. The periphery interaction and spread results are weak. It does not show a strong additional yield response beyond the common movement between core and peripheral yields.

This weaker spread result is not surprising. ECB communication affects expected European rates and therefore moves sovereign yields together. That means that a hawkish ECB surprise would result in a higher yield for both Germany and Italy because of the rise in interest rates. Therefore, even though both yields may respond strongly, the spread or difference between them may not change or change only slightly. The spread captures only the additional periphery-specific component after subtracting the core response. This is also consistent with the pooled panel results, where core/periphery interactions are not statistically significant.

As an additional sensitivity check, I also estimated this specification using only the pre-2019 sample. The results were largely similar to the full-sample estimates, indicating that the weak spread result was not uniquely driven by the post-2019 period.

The result must not be misconstrued to mean that peripheral sovereign risk is irrelevant, but that in these high-frequency press-conference regressions, the additional periphery-over-core responses are not estimated precisely enough to become the central result. This is compatible with the fact that ECB communication moves the common rate-expectations of the euro-area sovereign yields, whereas periphery-specific risk premia are more closely represented and explained by fiscal-vulnerability measures such as debt-to-GDP.

5.7 Period Specific Panel Results

Table 5.7. Panel baseline regressions by period: 10Y sovereign-yield responses

Variable	**2002-2009**	**2010-2018**	**2019-sample end**
Timing surprise	0.265***	0.342*	0.831*
SE	(0.0573)	(0.177)	(0.464)
Forward-guidance surprise	0.389***	0.459***	0.807***
SE	(0.0591)	(0.154)	(0.134)
Observations	360	368	220
R-squared	0.558	0.0911	0.363

The baseline transmission effect is positive in all three periods, with larger estimated coefficients after 2019.

According to the baseline period-split panel results, both timing and forward guidance surprises raise 10-year sovereign yields in all three periods. All forward guidance coefficients are positive and statistically significant. The same is also true for the timing coefficients except the one for 2019-2025 which is less precise. This is due to the fact that this period contains fewer ECB events.

The main takeaway from this section is that ECB press conference communication does matter throughout the sample, but the estimated response is higher after 2019. During the inflation surge and rising rate cycle, markets were extremely focused on the future path and persistence of ECB tightening. Therefore, as compared to earlier periods, in this period, timing and forward guidance carried more information for sovereign yield changes.

Tables 5.8 and 5.9 add VSTOXX and VIX interactions within the same period split. These tables are included as supporting evidence because splitting the sample reduces precision, especially in the later period. The main interest is therefore the broad sign and pattern of the interaction terms rather than every period-specific coefficient.

Table 5.8. Panel VSTOXX interaction regressions by period

Variable	2002-2009	2010-2018	2019-sample end
Timing surprise	0.288***	0.555***	0.837
SE	(0.0615)	(0.153)	(0.616)
Forward-guidance surprise	0.428***	0.582***	0.723***
SE	(0.0504)	(0.173)	(0.173)
VSTOXX, standardised	0.0222	0.775*	0.507
SE	(0.164)	(0.444)	(0.614)
Timing x VSTOXX	-0.0744	-0.612**	-1.499*
SE	(0.0509)	(0.251)	(0.793)
Forward guidance x VSTOXX	-0.159**	-0.214**	-0.165
SE	(0.0702)	(0.0871)	(0.253)
Observations	360	368	200
R-squared	0.618	0.127	0.421

The VSTOXX dampening pattern is strongest in the middle period; reduced sample sizes make later period estimates less precise.

Table 5.9. Panel VIX interaction regressions by period

Variable	2002-2009	2010-2018	2019-sample end
Timing surprise	0.310***	0.482***	1.146**
SE	(0.0582)	(0.144)	(0.474)
Forward-guidance surprise	0.477***	0.587***	0.789***
SE	(0.0414)	(0.176)	(0.128)
VIX, standardised	0.00363	0.701	0.426
SE	(0.138)	(0.430)	(0.510)
Timing x VIX	-0.0886*	-0.847***	-1.144*
SE	(0.0453)	(0.282)	(0.609)
Forward guidance x VIX	-0.187***	-0.229*	-0.0561
SE	(0.0463)	(0.117)	(0.165)
Observations	360	368	220
R-squared	0.667	0.138	0.429

The VIX period-split results broadly reinforce the uncertainty dampening interpretation.

The VIX period split results provide additional supportive evidence on the stability of the uncertainty-dampening mechanism across all regimes. The clearest VIX dampening evidence appears before the 2019 split, especially for forward guidance surprises.

5.8 Fiscal Vulnerability: Debt-to-GDP Extension

Table 5.10. Debt-to-GDP fiscal-vulnerability interaction

Variable	10Y sovereign-yield response
Timing surprise	0.355***
SE	(0.107)
Debt/GDP, standardised	-0.00500
SE	(0.220)
Timing x Debt/GDP	0.101
SE	(0.0799)
Forward-guidance surprise	0.513***
SE	(0.0629)
Forward guidance x Debt/GDP	0.0901**
SE	(0.0380)
Observations	948
R-squared	0.219

This extension uses debt-to-GDP as fiscal vulnerability, not as an uncertainty index. The forward guidance interaction is positive and significant.

The results show that the forward-guidance × debt/GDP interaction is positive and statistically significant while the timing × debt/GDP is positive but not statistically significant. This distinction helps explain why the amplification channel is not the central VSTOXX result. First, a broad uncertainty measure and a fiscal vulnerability measure do not test the same thing. Second, VSTOXX and the data used in this context reflect the market state before the ECB event whereas debt-to-GDP involves a country's exposure to future interest rates and refinancing pressure. And third, debt sustainability is more sensitive to forward guidance instead of timing surprises because debt financing is more sensitive to the expected path of rates over time since persistent higher rates affect refinancing costs and debt servicing burdens. Therefore, it is plausible that fiscal vulnerability does follow the amplification channel, and that amplification is clearer and relevant for forward guidance compared to timing surprises.

The results therefore complement the baseline VSTOXX results and outline how considering different state variables highlights two different mechanisms.

5.9 Equity Market Checks

The equity results are more mixed, and that is economically understandable. A hawkish ECB surprise can lower equity prices by increasing discount rates and diminishing expected growth. For bank equities, however, higher expected rates may improve net interest margins while also increasing credit risk, lowering loan demand, and reducing the value of existing bond portfolios. Therefore, there are multiple competing channels which make equity responses less direct as compared to sovereign-yield responses.

5.10 Summary of Results

Overall, the results confirm a state-dependent interpretation of ECB monetary policy transmission to euro area sovereign yields. The baseline country specific results demonstrate how press conference timing and forward guidance surprises raise sovereign yields and thereby indicate the basic communication channel: hawkish information regarding future path of policy is priced into government bond markets through higher expected future rates.

The state-dependent results then proceed to show that this response is dampened when market uncertainty or financial volatility is higher. Both VSTOXX and VIX interactions are negative in the main country level regressions and the pooled-country panel. The dampening result is therefore not confined to one country, one maturity, or one uncertainty measure. Along with this, the results do not support a simple claim that periphery yields always amplify relative to core yields under higher market uncertainty. Instead, the spread/periphery interaction evidence is weaker. Differences in market liquidity can also help explain why some sovereign yield responses do not show a stronger effect for peripheral countries.

The debt-to-GDP extension helps demonstrate a second mechanism. When fiscal uncertainty is considered the state variable, the forward guidance response is observed to be amplified. The overall empirical picture is therefore not confined to the fact that state dependence always works in one direction. This distinction is central to the dissertation's interpretation of state-dependent ECB monetary policy transmission. It also avoids forcing all results into a single mechanism by recognising that different state variables capture different dimensions of uncertainty.

The evidence suggests that the relevant state variable matters, market-wide uncertainty and stress weaken the effect of the communication channel, whereas fiscal uncertainty can strengthen the yield response to persistent future rate guidance.

6 Conclusion

This dissertation examined the state dependence of European sovereign yields on the transmission of ECB monetary policy surprises. According to the findings in this dissertation, it can be concluded that ECB communication is relevant in the context of sovereign yields, but its effects are not fixed across states. Similar policy surprises may be transmitted differently depending on the financial and fiscal environments in which they are received. As shown above, market wide uncertainty can be associated with a weaker yield response whereas fiscal vulnerability is associated with a stronger response to forward guidance surprises. Therefore, state dependence is present, but the propagation is not one directional.

This is the main contribution of this dissertation. A simple view of state dependence would ask whether monetary policy transmission becomes stronger or weaker in certain periods. But the results here indicate that this framing would be too narrow as even though market uncertainty and fiscal vulnerability are related, they are not the same. Market uncertainty weakens/dampens the immediate response because the ECB signal is received alongside other forces such as recession concerns, safe-haven concerns, liquidity conditions, policy reversal expectations and inflation expectations effects. Fiscal vulnerability on the other hand works through a different channel called amplification because persistent future rates guidance has direct implications for refinancing costs, debt sustainability and risk premia. For a monetary union such as the euro area, this distinction is quite important since ECB provides a common policy signal, but sovereign yields reflect country specific fiscal exposure, liquidity conditions and risk premia. As a result, the direction of state dependence depends on what the state variable is measuring.

Certain limitations exist due to the nature of the analysis, and they must be mentioned. First, even though the high frequency design is useful for isolating the immediate market response around ECB announcements, it does not measure long-run macroeconomic effects of ECB communication. Second, the uncertainty variables are useful, but imperfect proxies, and must therefore be interpreted accordingly. And third, debt-to-GDP is a slow-moving measure of fiscal uncertainty and cannot be interpreted as a corresponding high frequency market stress measure. Some extensions in the analysis rely on smaller subsamples and therefore the results are interpreted with appropriate caution.

Future research could build on this dissertation by linking high frequency ECB surprise measure to lower frequency macro financial outcomes, for example through local projections or proxy-SVAR methods. This would enable further analysis to determine whether the immediate yields responses identified here persist in the event window and thereby affect broader financing conditions. Further work could also separate the roles of liquidity, inflation expectations and sovereign risk premia more directly in the transmission process.

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8 Appendix

Some of the tables, graphs, and formatting in this dissertation were created or edited with the assistance of OpenAI ChatGPT. It was used as a tool to organise and present material, but the underlying data, empirical results, interpretation, and final academic judgement are my own.

Note: In this appendix, I report supporting outputs that are relevant to the dissertation but do not need to be placed in the main text. The main empirical discussion should remain focused on the press-conference timing and forward-guidance results, while these tables provide additional transparency around summary statistics, press release target surprise checks and equity market checks.

For regression tables, standard errors are reported in parentheses. *, ** and *** denote statistical significance at the 10%, 5% and 1% levels, respectively.

Appendix A. Additional Summary Statistics

Table A1. Summary statistics for the press conference sample

	count	**mean**	**sd**	**min**	**max**
Germany 10Y	237	-0.009	3.203	-12.650	12.465
France 10Y	237	0.021	3.313	-14.050	14.285
Italy 10Y	237	0.355	5.332	-14.000	38.000
Spain 10Y	237	0.100	4.344	-12.050	30.950
Timing surprise	237	0.000	2.029	-11.670	10.276
Forward-guidance surprise	237	0.000	3.624	-26.044	14.738
QE surprise	99	-0.004	2.711	-9.228	7.556
VSTOXX	232	23.093	9.052	12.240	58.204
VSTOXX, standardised	232	0.000	1.000	-1.199	3.879

Table A2. Summary statistics for the press release sample

	count	mean	sd	min	max
Germany 10Y	242	-0.123	1.923	-9.600	8.200
France 10Y	242	-0.185	2.340	-10.800	10.650
Italy 10Y	242	-0.395	3.591	-19.150	18.000
Spain 10Y	242	-0.299	2.619	-11.200	11.850
Target surprise	242	0.000	2.639	-13.661	21.122
VSTOXX	237	23.171	9.140	12.240	58.204
VSTOXX, standardised	237	-0.000	1.000	-1.196	3.833

Appendix B. Press Release Target Surprise Checks

These tables report supplementary checks using the press release target surprise. They are included as a preliminary comparison because the main dissertation focuses on press conference timing and forward guidance surprises.

Table B1. Press release baseline regressions

	(1)	(2)	(3)	(4)
	Germany	**France**	**Italy**	**Spain**
Target surprise	-0.0693	-0.0120	0.0153	0.0441
	(0.0837)	(0.0985)	(0.130)	(0.120)
Constant	-0.123	-0.185	-0.395*	-0.299*
	(0.123)	(0.151)	(0.231)	(0.169)
Observations	242	242	242	242
R-squared	0.00904	0.000183	0.000126	0.00197

Table B2. Press release VSTOXX interaction regressions

	(1)	(2)	(3)	(4)
	Germany	**France**	**Italy**	**Spain**
Target surprise	-0.123	-0.194	-0.272	-0.0596
	(0.138)	(0.153)	(0.231)	(0.217)
VSTOXX, standardised	0.138	0.201	0.463*	0.281
	(0.145)	(0.180)	(0.249)	(0.182)
Target surprise x VSTOXX	0.0281	0.121	0.183	0.0567
	(0.0573)	(0.0757)	(0.124)	(0.0961)
Constant	-0.115	-0.232	-0.478**	-0.309*
	(0.130)	(0.153)	(0.234)	(0.170)
Observations	237	237	237	237
R-squared	0.0181	0.0374	0.0464	0.0189

Table B3. Press release high uncertainty regressions

	(1)	(2)	(3)	(4)
	Germany	**France**	**Italy**	**Spain**
Target surprise	-0.126	-0.169	-0.258	0.0627
	(0.191)	(0.230)	(0.401)	(0.332)
High VSTOXX=1	0.256	0.180	0.302	0.212
	(0.321)	(0.393)	(0.625)	(0.441)
High VSTOXX=1 # Target surprise	0.0563	0.190	0.335	-0.0399
	(0.216)	(0.260)	(0.426)	(0.353)
Constant	-0.177	-0.254	-0.532**	-0.319
	(0.145)	(0.176)	(0.267)	(0.196)
Observations	237	237	237	237
R-squared	0.0145	0.00923	0.0116	0.00309

Appendix C. Equity Market Checks

These tables report supporting equity market checks. They are included because equity responses provide an additional financial-market comparison, but the sovereign-yield results remain the focus of the dissertation.

Table C1. Equity market press-release baseline checks

	(1)	(2)
	STOXX50	**Bank equities**
Target surprise	-0.0320**	-0.0182
	(0.0159)	(0.0189)
Constant	-0.00921	0.0247
	(0.0215)	(0.0383)
Observations	242	242
R-squared	0.0605	0.00646

Table C2. Equity market press release VSTOXX checks

	(1)	(2)
	STOXX50	**Bank equities**
Target surprise	0.0162	0.0529**
	(0.0162)	(0.0263)
VSTOXX, standardised	-0.0925***	-0.0841**
	(0.0236)	(0.0337)
Target surprise x VSTOXX	-0.0303***	-0.0481***
	(0.0107)	(0.0140)
Constant	0.00855	0.0534
	(0.0217)	(0.0415)
Observations	237	237
R-squared	0.222	0.0973

Table C3. Equity market press conference baseline checks

	(1)	(2)
	STOXX50	**Bank equities**
Timing surprise	-0.00164	0.00652
	(0.0196)	(0.0260)
Forward-guidance surprise	-0.0260*	-0.0223
	(0.0132)	(0.0216)
Constant	-0.0961**	-0.191***
	(0.0382)	(0.0677)
Observations	237	237
R-squared	0.0253	0.00617

Table C4. Equity-market press conference VSTOXX checks

	(1)	(2)
	STOXX50	**Bank equities**
Timing surprise	-0.0229	-0.0152
	(0.0174)	(0.0237)
Forward-guidance surprise	-0.0316**	-0.0266
	(0.0135)	(0.0210)
VSTOXX, standardised	-0.0580	-0.119
	(0.0442)	(0.0917)
Timing surprise x VSTOXX	0.0503***	0.0466
	(0.0160)	(0.0302)
Forward guidance x VSTOXX	0.0294**	0.00302
	(0.0136)	(0.0244)
Constant	-0.0953**	-0.197***
	(0.0381)	(0.0686)
Observations	232	232
R-squared	0.0763	0.0284

Appendix D. Additional Data Cleaning Details

D1. VIX Data Cleaning

The VIX data were downloaded from FRED as a daily CSV series. The raw FRED file was imported into Stata, the date variable was converted into Stata date format, the VIX values were converted to numeric format where necessary, missing values were dropped, and duplicate dates were removed.

D2. Debt-to-GDP Data Cleaning

The original Eurostat spreadsheet was imported into Stata, and the data were reshaped from a wide quarterly format into a long country-quarter format.

Country names were converted into country codes matching the event level sovereign yield dataset. The raw debt-to-GDP values were converted into numeric format. Quarterly date variables were then created from the original quarter labels. For each ECB event, debt-to-GDP was matched using the previous quarter's value. For example, an ECB event occurring in a given quarter is matched to the debt-to-GDP value from the preceding quarter. This avoids using fiscal information that would not have been available at the time of the ECB event. The cleaned debt-to-GDP variable is merged into the long press conference country-event dataset.